

Curriculum Vitae



Personal Information

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City: Berlin



Education

2024.9 - present Technische Universität Berlin / M.Sc Fahrzeugtechnik

2018.9 - 2022.6 Beijing Forestry University / B.Eng. Fahrzeugtechnik

Technical Skills and Tools:

8051 Microcontroller, MATLAB/Simulink, ROS1/ROS2, Git, Docker, Siemens NX, Solid Edge, CAD/CAE, Gmsh, Actran, Python, C, Linux (Kali/Ubuntu), MySQL, Flask, Django, HTML/CSS/JavaScript, Android Studio, Stable Diffusion, Trae/Codex

Computer Science: Deep Learning, Machine Learning, Digital Image Processing, Software Engineering, Embedded Systems, Python, C Programming, Computer Architecture, Computer Networks, Algorithms and Databases, Full-Stack Development

Automotive Engineering: Autonomous Driving, CAD in Automotive and Mechanical Engineering, Microcontrollers, Vehicle Transmission Technology, Sustainable Internal Combustion Engine Powertrains, Sustainable Electrochemical Vehicle Powertrains, Mechatronics, MATLAB/Simulink

Project Experience

2025.3-2026.3 Optimization of Path Planning for Autonomous Model Vehicle (VDI)

Development and improvement of a local path planning module using ROS2 and Python. Optimization of the previous fixed-offset approach through a dynamic lane-offset strategy for curve driving. Estimation of lane curvature using KASA circle approximation, smoothing of offset changes with a sliding-window approach, and validation through Python simulations, real track maps and ROS2 vehicle tests

2025.3-2026.3 Development of Image Processing for Autonomous Model Vehicle (VDI)

Development of a camera-based perception module using OpenCV, YOLOv8 and ROS2. Implementation of BEV-based lane detection with ROI selection, perspective transformation, color-space segmentation and morphological image processing. Annotation of traffic sign and traffic light data with Roboflow, training of a YOLOv8 model, ONNX export and deployment on the ADXA platform with real-time visualization of bounding boxes, classes, confidence scores and inference latency

2022.1-2022.6 Investigation of the Mechanical Properties of PMMA under Heating Conditions

Development of a one-dimensional steady-state heat conduction model in MATLAB to calculate the temperature distribution in PMMA. Conducted three-point bending tests under heating conditions and evaluated force-time data at different temperatures and material thicknesses. Analyzed resistance drop, peak force, steady-state force, elastic modulus and maximum bending stress to determine suitable thermoforming temperatures for 5 mm and 8 mm PMMA in the range of 60-95 °C

Additional Information

Languages: Chinese native speaker, German DSH-2, good English skills

Competition: VDI Autonomous Driving Challenge 2025, Team smart TUBe_Rookies